

xrpl_agent_id

EXECUTIVE SUMMARY

Portable, vendor-independent identity for AI agents
XRPL native · MIT licensed

Identity & Trust · XRP Ledger (XLS-70 Credentials + XLS-40d DIDs)
v0.4.0 · **Credential expiry enforcement** · 201 tests passing · Live on XRPL Testnet

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Today, agent identity is a promise. This makes it a public fact.

THE PROBLEM

AI agents are taking real actions – spending money, calling APIs, signing contracts. But they have **no public identity** and **no portable way to prove who vouched for them**. Today, agent trust is a promise made by whoever runs the platform. That trust is invisible, vendor-locked, and impossible to audit. When an agent misbehaves, there's no public trail – only the platform vendor's private logs.

THE SOLUTION

`xrpl_agent_id` – a Python library that gives every AI agent a public, verifiable identity on the XRP Ledger. **No new chain. No new consensus rule. No validator vote.** Just the existing XLS-70 Credential and XLS-40d DID primitives, packaged for AI developers. Anyone in the world can verify any agent's identity and credentials with a single `ledger_entry` call – no API key, no permission, no trust in a vendor's internal logs.

HOW IT WORKS

Three on-ledger transactions:

1. **Issuer** signs `CredentialCreate` – public attestation ("this agent passed audit X")
2. **Agent** signs `CredentialAccept` – opts in; acceptance itself becomes auditable on-ledger
3. **Agent** signs `DIDSet` – pins its W3C-compatible DID Document (URI-only pattern, fits the 256-byte on-ledger cap)

Verifiers query the ledger. They see the credential, the issuer, the timestamp, the URI to the full claim details. Done.

WHAT'S ALREADY BUILT (LIVE DEMO)

Subject:	<code>CredentialCreate: 3C03A3E9...59E7 (lgr 21123060)</code>
<code>did:xrpl:2:rPNGAyKowBrtpbeVjzDkHG4uJBKdV1q3cf</code>	<code>CredentialAccept: 580B8CD7...243A (lgr 21123062)</code>
Issuer:	<code>DIDSet: A230469E...A3B3 (lgr 21123064)</code>
<code>did:xrpl:2:rHdfNdLUEd7tBXZCaUuuK5wMcKzhtpNDBU</code>	

Library state: Full Python package, MIT licensed, **v0.4.0**. **201 offline tests passing, 6 live integration tests** (XRPL testnet; credential expiry, DID roundtrip, authorization policy). **v0.4.0 closed a silent correctness gap:** `CREDENTIAL_EXPIRED` is now emitted when an XLS-70 credential's `expiration` is in the past. Ergonomic API: `AgentIdentity.from_seed()`, `issuer.issue_credential()`, `agent.has_credential()`, `resolve_did()`.

WHY XRPL

- **Both primitives already live on mainnet** – XLS-40d since 2022, XLS-70 since 2024. Zero protocol changes required.
- **3–5 second finality** – fast enough for agent-to-agent commerce
- **Sub-cent fees** – cheap enough for high-frequency agent activity
- **Native hooks for future enhancement** – credential-aware transactions, native DID resolution RPC, MPC signers are all on the roadmap without requiring forks

WHY NOW

Agent-to-agent commerce is the next frontier – LangChain, MCP, Composio are all exploding. Every agent framework today inherits the same closed-trust problem. **The window for setting the standard is open – the first credible Python library wins.**

THE ASK

Looking for three things:

1. **Validation** from the XRPL community on the credential-type taxonomy (`KYC`, `EVAL`, `OPERATOR`, `COMPLIANCE`, etc.)
2. **An early issuer partner** to run a pilot – any organization willing to issue credentials against real audit data

